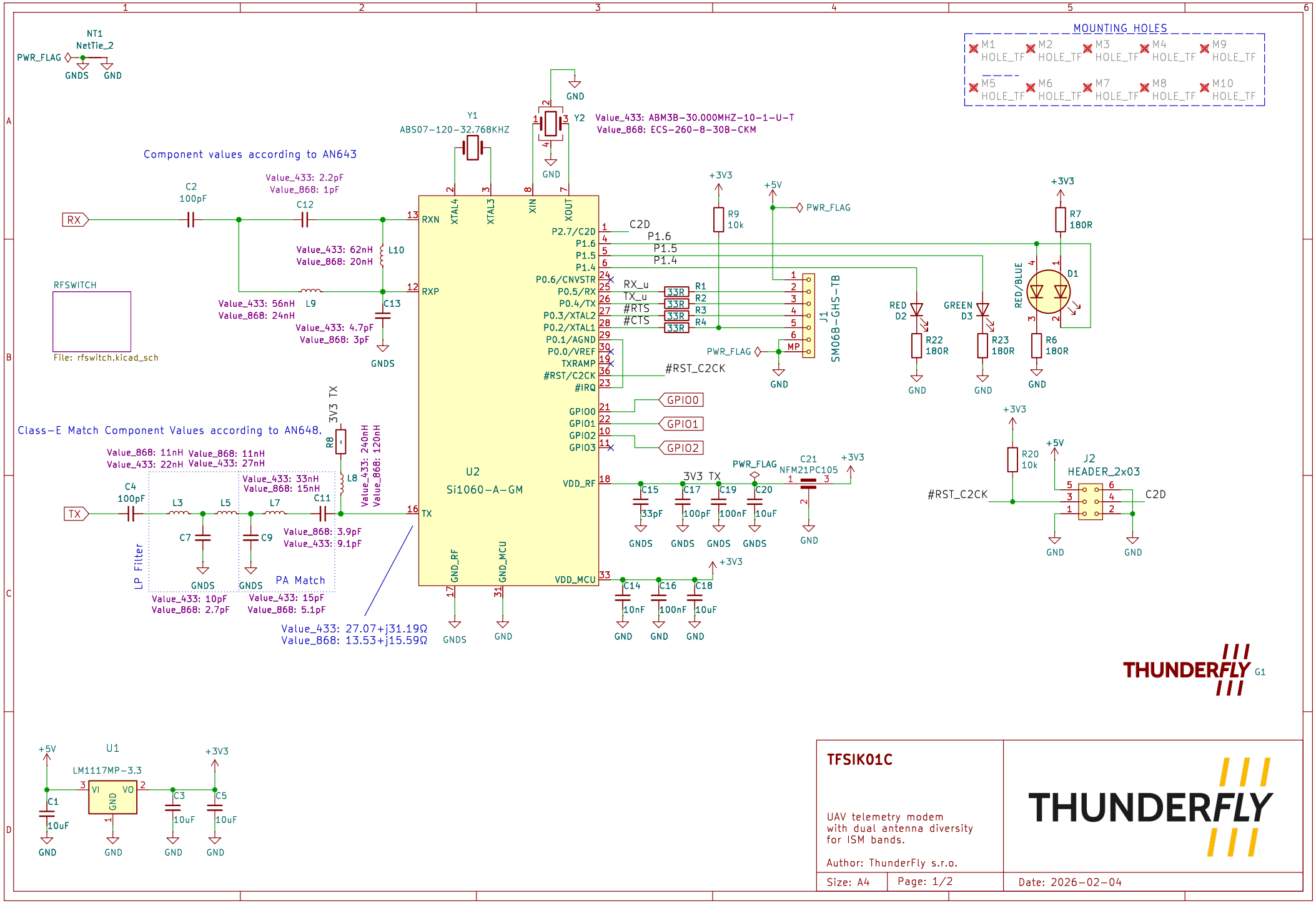
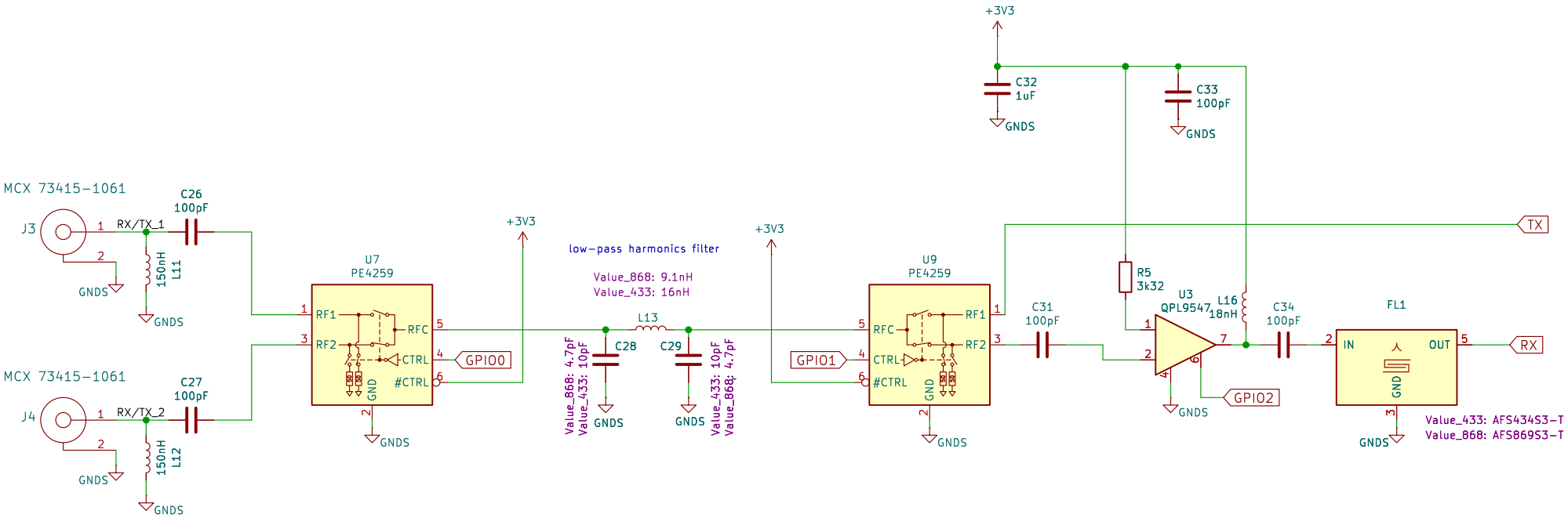

MOUNTING HOLES
M1 HOLE_TF
M2 HOLE_TF
M3 HOLE_TF
M4 HOLE_TF
M9 HOLE_TF
M5 HOLE_TF
M6 HOLE_TF
M7 HOLE_TF
M8 HOLE_TF
M10 HOLE_TF
NT1
NetTie_2
PWR_FLAG
GNDS
GND
Y1
ABS07-120-32,768KHZ
Y2
Value_433: ABM3B-30.000MHZ-10-1-U-T
Value_868: ECS-260-8-30B-CKM
Component values according to AN643
RX
C2
100pF
Value_433: 2.2pF
Value_868: 1pF
C12
Value_433: 62nH
Value_868: 20nH
L10
Value_433: 56nH
Value_868: 24nH
L9
C13
Value_433: 4.7pF
Value_868: 3pF
RFSWITCH
File: rfswitch.kicad_sch
XTAL4
XTAL3
XIN
XOUT
RXN
RXP
P2.7/C2D
P1.6
P1.5
P1.4
P0.6/CNVSTR
P0.5/RX
P0.4/TX
P0.3/XTAL2
P0.2/XTAL1
P0.1/AGND
P0.0/VREF
TXRAMP
#RST/C2CK
#IRQ
GPIO0
GPIO1
GPIO2
GPIO3
VDD_RF
VDD_MCU
GND_RF
GND_MCU
TX
U2
Si1060-A-GM
C2D
RX_u
TX_u
#RTS
#CTS
33R
R1
R2
R3
R4
J1
SM06B-GHS-TB
MP
#RST_C2CK
+3V3
+5V
R9
10k
R7
180R
RED
D2
GREEN
D3
RED/BLUE
D1
R22
180R
R23
180R
R6
180R
3V3 TX
R8
Class-E Match Component Values according to AN648.
Value_868: 11nH
Value_433: 22nH
Value_868: 11nH
Value_433: 27nH
Value_433: 33nH
Value_868: 15nH
Value_433: 240nH
Value_868: 120nH
L8
C4
100pF
L3
L5
L7
C11
C7
C9
Value_868: 3.9pF
Value_433: 9.1pF
LP Filter
PA Match
Value_433: 10pF
Value_868: 2.7pF
Value_433: 15pF
Value_868: 5.1pF
Value_433: 27.07+j31.19Ω
Value_868: 13.53+j15.59Ω
C15
33pF
C17
100pF
C19
100nF
C20
10uF
C21
NFM21PC105
C14
10nF
C16
100nF
C18
10uF
R20
10k
J2
HEADER_2x03
U1
LM1117MP-3.3
VI
VO
C1
10uF
C3
10uF
C5
10uF
THUNDERFLY
G1
TFSIK01C
UAV telemetry modem
with dual antenna diversity
for ISM bands.
Author: ThunderFly s.r.o.
Size: A4
Page: 1/2
Date: 2026-02-04



TFSIK01C

UAV telemetry modem
with dual antenna diversity
for ISM bands.

Author: ThunderFly s.r.o.

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THUNDERFLY